

Complaint Management System - Project Documentation

1. Project Overview

The Complaint Management System is a full-stack MERN (MongoDB, Express, React, Node.js) application designed to bridge the gap between citizens and government departments. It allows users to report issues like water leakage, electricity failures, or road damage and track their progress in real-time.

2. Technology Stack

- Frontend: React.js, Tailwind CSS, Axios, React Router
- Backend: Node.js, Express.js
- Database: MongoDB (with Mongoose ODM)
- Authentication: JSON Web Tokens (JWT), Bcrypt.js for password hashing

3. Key Features

- User Side: Secure registration/login, complaint submission with category selection, and a personal dashboard to track status.
- Admin Side: Administrative dashboard to view all complaints, filter by category, and update status (Pending, In Progress, Resolved).
- Security: Role-based access control (RBAC) and protected API endpoints.

4. Installation & Setup

1. Prerequisites: Node.js and MongoDB installed.
2. Backend Setup:
 - Navigate to /server folder.
 - Run 'npm install'.
 - Create a .env file with PORT, MONGODB_URI, and JWT_SECRET.
 - Run 'npm start' or 'npm run dev'.
3. Frontend Setup:
 - Navigate to /client folder.
 - Run 'npm install'.
 - Run 'npm run dev' to start the Vite/React development server.

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5. Project Structure

server/

- models/: User and Complaint schemas.
- controllers/: Business logic for auth, complaints, and admin.
- routes/: API endpoint definitions.
- middleware/: JWT authentication and role checking.

client/

- src/pages/: Login, Register, UserDashboard, AdminDashboard.
- src/components/: Navbar, ComplaintForm, ComplaintCard.
- src/context/: AuthContext for global state.

6. API Endpoints

- POST /api/auth/register: User registration.
- POST /api/auth/login: User/Admin login.
- GET /api/complaints: Fetch complaints for logged-in user.
- POST /api/complaints: Submit a new complaint.
- GET /api/admin/all: Fetch all complaints (Admin only).
- PATCH /api/admin/status/:id: Update complaint status (Admin only).